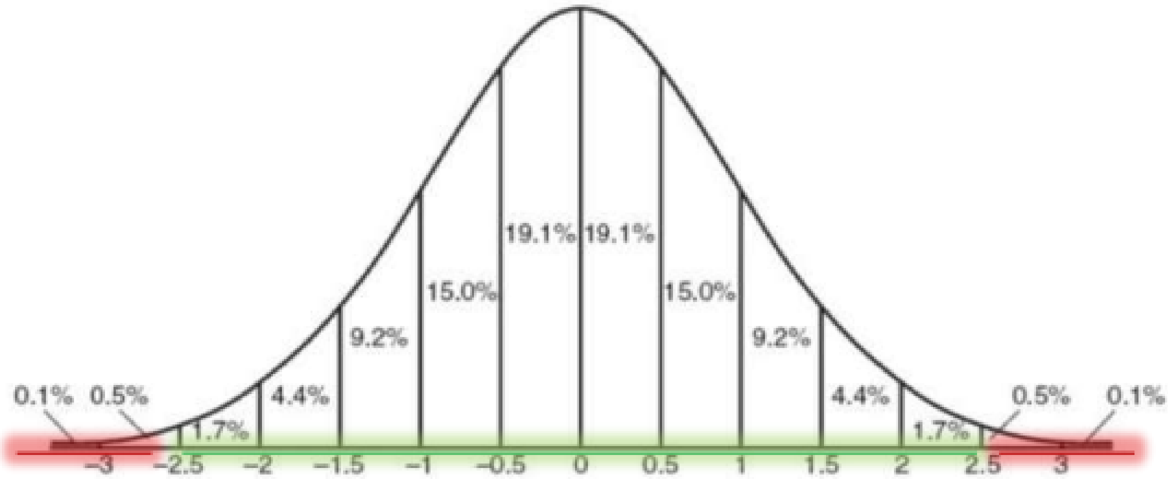

MASTER 1 2023-2024 RESEARCH PROJECT REPORT

EXTREME VALUE THEORY FOR INSURANCE

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Contents

Abstract

Our project delves deeply into the theory of extreme values and its practical applications, particularly in the domain of insurance. We first explore fundamental approaches such as the block maxima or minima method and the threshold exceedance approach based on the Generalized Pareto Distribution. Secondly, we propose different methods to estimate reinsurance premiums and the costs of covering extreme risks, thereby providing valuable tools for pricing these rare and critical phenomena while suggesting advanced estimation methods to assist in risk management. Finally, we apply all the theory to a real dataset.

Introduction

The Extreme Values Theory (EVT) is a branch of statistics that emerged between 1920 and 1943, thanks to Fréchet (1927), Fisher and Tippet (1928), and Gnedenko (1943). This branch has experienced significant growth since the 1950s, although the main theorem was stated by Fisher and Tippet as early as 1928. Its purpose is to model and describe the occurrence and intensity of so-called extreme events, which exhibit very large variations (with a low probability of occurrence). When the behavior of these events is due to randomness, their distribution can be studied. They are considered extreme when they are significantly larger or smaller than the values usually observed.

The extreme values theory is a crucial tool in insurance. Indeed, in the field of insurance, the pricing of policies is often hindered by a crucial lack of information regarding extreme events, especially when it comes to statistically analyzing claims that significantly deviate from the average. Moreover, data on the most frequent claims tends to overshadow those related to extreme events, which can lead to gaps or unreliable information in risk assessment.

In this paper, we will present the probabilistic and numerical tools for Extreme Value Theory and we will implement these methods in a case study.

I Presentation and overview of extreme value theory

In Extreme Value Theory, two main approaches are distinguished. The first approach concerns the Generalized Extreme Value (GEV) distribution and focuses on the distribution of the maximum or minimum suitably normalized. This first approach is based on the theorem of Fisher and Tippet. The second approach concerns the distribution of excesses. It involves using observations that exceed a certain deterministic threshold. The differences between these observations and the threshold are called excesses. This approach is based on the convergence towards a Generalized Pareto Distribution (GPD).

1.1 Block Maxima or Block Minima Approach

In this approach, we group observations into fixed-sized blocks and focus on the distribution of maxima (minima) within each block for a Generalized Extreme Value (GEV) distribution. It is necessary to find a compromise between the block size, which must be large enough for the GEV distribution approximation to be realistic, and the number of blocks, which must be large enough to obtain precise estimates of the three GEV parameters. This method has the well-known drawback of not utilizing all the information present in the data on extreme events since it retains only one value per block for statistical inference.

1.1.1 Asymptotic Behavior of Extremes

Let $X_1, \ldots, X_n$ be a sequence of independent and identically distributed random variables with distribution function F, and let $M_n = \max(X_1, \ldots, X_n)$.

In extreme value theory, the aim is to determine the distribution followed by the maximum or minimum based on the distribution of the random variable X.

To do this, we calculate the distribution function:

$$\begin{aligned} F_{M_n}(x) &= P(M_n \leq x) \\ &= P(X_1, \ldots, X_n \leq x) \\ &= P(X_1 \leq x) \ldots P(X_n \leq x) \\ &= (F_X(x))^n. \end{aligned}$$

From these results, we draw the conclusion that the maximum M_n is a random variable whose distribution function corresponds to F^n. Since the distribution function of X is often not known, it is generally not possible to determine the distribution of the maximum from this result. Moreover, we often focus on M_n when the sample size is large, and we aim to have asymptotic approximations.

$$\lim_{n\to\infty} F_{M_n}(x) = \lim_{n\to\infty} F_X(x)^n = \begin{cases} 0 & \text{if } F(x) < 1 \\ 1 & \text{if } F(x) = 1 \end{cases}$$

We observe that the asymptotic distribution of the maximum, determined by letting n tend to infinity, results in a degenerate distribution.
As the distribution function obtained previously leads to a degenerate distribution when n tends to infinity, we search for a non-degenerate distribution for the maximum of X. This non-degenerate limiting distribution is provided by the 'Extreme Value Distribution Theorem,' which gives a necessary and sufficient condition for the existence of a non-degenerate limiting distribution for the maximum. This theorem was proposed by Gnedenko (1943), who provided the forms of the limiting distributions, and Jenkinson (1955), who provided the general expression for them.

Theorem (Fisher-Tippett) : If there exist sequences of real numbers $c_n > 0$ and d_n, such that when $n \to \infty$ we have $P((M_n - d_n)/c_n \leq x) \longrightarrow G(x)$ for a non-degenerate distribution G, then G is of the same type as one of the following three distributions:

$$G_0(x) = \exp(-e^{-x}) \quad \text{for } x \in \mathbb{R}$$

$$G_{1,\alpha}(x) = \begin{cases} 0 & \text{if } x < 0 \\ \exp(-x^{-\alpha}) & \text{if } x \geq 0 \text{ and } \alpha > 0 \end{cases}$$

$$G_{2,\alpha}(x) = \begin{cases} \exp(-(-x)^{-\alpha}) & \text{if } x < 0, \alpha < 0 \\ 1 & \text{if } x \geq 0 \end{cases}$$

We call G_0 the Gumbel distribution, $G_{1,\alpha}$ the Fréchet distribution, and $G_{2,\alpha}$ the Weibull distribution. These three probability laws are called extreme value distributions

In some cases, it is preferable to have a single distribution that unifies the three previous ones. Von Mises (1954) and Jenkinson (1955) proposed the Generalized Extreme Value (GEV) distribution, denoted by GEV(μ, σ, ξ).

$$G_{\mu,\sigma,\xi}(x) = \begin{cases} \exp\left(-\left(1 + \xi\frac{(x-\mu)}{\sigma}\right)^{-\frac{1}{\xi}}\right) & \text{if } \xi \neq 0 \\ \exp\left(-\exp\left(-\frac{(x-\mu)}{\sigma}\right)\right) & \text{if } \xi = 0 \end{cases}$$

Where ξ is a shape parameter, also called the index of extremes or tail index, μ is a location parameter and σ is a scale parameter. The unification of the standard generalized extreme value distribution law into a single distribution function facilitates the study of the behavior of the maximum or minimum. Furthermore, for $\mu = 0$ and $\sigma = 1$, we obtain the standard form of the three types of extreme value distributions. This distribution depends only on the shape parameter ξ. Thus, $G_{\mu,\sigma,\xi}(x)$ becomes:

$$G_\xi(x) = \begin{cases} \exp\left(-(1 + \xi x)^{-\frac{1}{\xi}}\right) & \text{if } \xi \neq 0, \text{ with } (1 + \xi x) > 0, \\ \exp(-\exp(-x)) & \text{if } \xi = 0. \end{cases}$$

The larger the absolute value of ξ, the more weight the extremes have in the original distribution. This leads to what is known as 'heavy-tailed' distributions. Three cases are possible:

1. $\xi < 0$: G follows the Weibull distribution.

2. $\xi > 0$: G follows the Fréchet distribution.

3. $\xi \to 0$: G follows the Gumbel distribution.

In addition, in EVT, the most important information is found in the tail of the distribution and is characterized by the shape parameter ξ. Indeed, depending on its sign, we distinguish three domains of attraction that will be presented later.

1.1.2 Domains of Attraction of GEV Distributions

The theorem of convergence of the limit distribution of the maximum states that if $P((M_n - b_n)/a_n \leq x) = (F(a_n x + b_n))^n \longrightarrow G(x)$ where G is a non-degenerate distribution, then G has a generalized extreme value distribution.
Definition: We say that F belongs to the domain of attraction of G ($F \in D(G)$) if there exist two sequences (a_n) and (b_n) such that the previous convergence occurs.

Definition: A function L is said to be slowly varying if $L(t) > 0$ for t sufficiently large, and for all $x > 0$, we have:

$$\lim_{t\to\infty} \frac{L(tx)}{L(t)} = 1$$

Domain of attraction of the Gumbel distribution:

The distribution exhibits exponential decay in the tail, which allows to characterize in this case, the domain of attraction of Gumbel denoted $D(\Lambda)$.

If there exists a measurable function R, called an auxiliary function, such that:

$$\lim_{x\to x_F} \frac{1 - F(t + xR(t))}{1 - F(t)} = \exp(-x)$$

where $x_F = \sup\{x \in \mathbb{R} \,|\, F(x) < 1\}$ is the right endpoint of F, then $F \in D(\Lambda)$.
Example: The normal, exponential, log-normal, and Weibull distributions belong to the domain of attraction of the Gumbel distribution.

Domain of attraction of the Weibull distribution:
The distributions in this domain are bounded on the right, and therefore, the right endpoint x_F is finite; the domain of attraction of the Weibull distribution is denoted $D(\Psi_\xi)$.

A distribution function F belongs to the domain of attraction $D(\Psi_\xi)$ if and only if $x_F < +\infty$ and

$$\overline{F}(x_F - \frac{1}{x}) = x^{-1/\xi} L(x)$$

where $\overline{F}$ is the survival function given by $\overline{F}(x). =$ 1 - F(x) , and L is a slowly varying function.
Example: The Uniform, Reverse Burr, Beta distributions belong to the domain of attraction of Weibull.

Domain of attraction of the Fréchet distribution:
The distributions belonging to this domain of attraction are characterized by a slow (polynomial) decay in the tail at infinity, with a right endpoint $x_F = +\infty$. the domain of attraction of the Weibull distribution is denoted $\mathrm{D}(\Phi_\xi)$

A distribution function F belongs to the domain of attraction $D(\Phi_\xi)$ if and only if its survival function is given by:

$$\overline{F}(x) = x^{-1/\xi} L(x)$$

where L is a slowly varying function.

Example: The Cauchy, Generalized Pareto, Student, Log-gamma distributions belong to the domain of attraction of Fréchet.

1.1.3 Estimation Methods for GEV Distribution Parameters

The GEV distributions are fundamentally distinguished by the parameter ξ. Accordingly, three cases are distinguished in parameter estimation. The first case corresponds to the Gumbel distribution for $\xi = 0$, the second corresponds to the Weibull distribution for $\xi < 0$, and the third corresponds to the Fréchet distribution for $\xi > 0$.

1.1.3.1 Estimation Methods for Gumbel Domain of Attraction Distributions

Maximum Likelihood Estimation

This method involves finding the parameters $\omega = (\mu, \sigma)$ that maximize the likelihood function of a sample of size n:

$$l(x_1, \ldots, x_n, \mu, \sigma) = \prod_{i=1}^{n} f(x_i, \mu, \sigma)$$

Consequently, its log-likelihood function is:

$$L_{MV}(\omega) = \log l(\omega)$$

By solving the following system representing the necessary optimality conditions:

$$\begin{cases} \frac{\partial L_{MV}(\omega)}{\partial \mu} = 0 \\ \frac{\partial L_{MV}(\omega)}{\partial \sigma} = 0 \end{cases}$$

we obtain an estimation of the parameters $= (\mu, \sigma)$. However, the solution is not explicit and requires numerical methods such as the Newton-Raphson method for resolution.

Method of Moments for the Gumbel Distribution

This method involves associating the first two moments, m_1 and m_2, respectively with the sample mean $(\bar{x}_n)$ and the empirical variance S_X^2.

In the case of a random variable following a Gumbel distribution, the method of moments leads to the following system:

$$\begin{cases} E(X) = \mu + \gamma\sigma \\ V(X) = \frac{1}{6}\pi^2\sigma \end{cases}$$

where $\gamma \approx 0.57721$ represents the Euler's constant. Solving this system of equations allows us to obtain the estimated values $\hat{\mu}$ and $\hat{\sigma}$:

$$\begin{cases} \hat{\sigma} = \frac{\sqrt{6}S_X}{\pi} \\ \hat{\mu} = \bar{x}_n - \gamma\hat{\sigma} \end{cases}$$

1.1.3.2 Parameter Estimation Methods for the Fréchet and Weibull Domain of Attraction

There are several parameter estimation methods for the Fréchet and Weibull distributions, such as the maximum likelihood method, the method of moments, the method of weighted probability moments, etc. But in this section, we present the maximum likelihood method which is common to both domains of attraction.

Maximum Likelihood Method

The method involves seeking the parameters $\theta = (\mu, \sigma, \xi)$ that maximize the log-likelihood function. Maximization leads to solving the following system of equations:

$$\begin{cases} \frac{\partial L_{MV}(\theta)}{\partial \mu} = 0 \\ \frac{\partial L_{MV}(\theta)}{\partial \sigma} = 0 \\ \frac{\partial L_{MV}(\theta)}{\partial \xi} = 0 \end{cases}$$

The issues related to estimation using the Maximum Likelihood Method have been studied by Smith. The obtained results are:
1. If $\xi > -0.5$, then the maximum likelihood estimates do not possess asymptotic properties such as convergence to the true value of the unknown parameter, invariance under parametric transformation, and asymptotic efficiency.
2. If $-1 < \xi < -0.5$, then the maximum likelihood estimators do not possess standard asymptotic properties.
3. If $\xi < -1$, then obtaining the maximum likelihood estimators is not guaranteed.

1.2 Threshold Exceedance Approach: GPD Law

The approach of block maxima often leads to information loss when selecting the maxima or minima within blocks. Taking this limitation into account has led to a new approach called the Peaks Over Threshold (POT) method. The approach by threshold exceedances, relies on the use of higher order statistics of the sample. It consists of retaining only the observations exceeding a certain threshold. The excess above the threshold is defined as the difference between the observation and the threshold.

Consider a sample of i.i.d random variables $Y_1, \ldots, Y_n$. Let u be a fixed threshold such that $u < y_F$ and the N_u observations $Y_{i1}, \ldots, Y_{iN_u}$ exceeding the threshold u. We call the excess above the threshold u the Z_j defined by $Z_j = Y_{ij} - u$, for $j = 1, \ldots, N_u$.

1.2.1 Modeling Excesses

Contrary to the block maxima approach, the POT method consists of using all observations, called excesses, that exceed a sufficiently high threshold. The goal is to analyze their asymptotic behavior. The idea of using an increasing number of order statistics from the sample was then further developed within the framework of the POT approach, through the approximation of the distribution of exceedances above a threshold by Generalized Pareto Distribution (GPD). More precisely, let $u < y_F$ and F_u be the distribution function of the exceedances defined by

$$F_u(y) = \mathrm{P}(Z \leq y | Y > u) = \mathrm{P}(Y - u \leq y | Y > u) = \frac{F(u+y) - F(u)}{1 - F(u)},$$

If F belongs to one of the three domains of attraction of the extreme value distributions (Fréchet, Gumbel, or Weibull), then there exists a strictly positive function σ_u and a $\xi \in \mathbb{R}$ such that

$$\lim_{u \to y_F} \sup_{y \in]0, y_F - u[} |F_u(y) - G_{\xi,\sigma_u}(y)| = 0;$$

where $y_F = \sup\{y \in \mathbb{R}, F(y) < 1\}$ is the right endpoint of F, and G_{ξ,σ_u} is the distribution function of the Generalized Pareto Distribution defined by:

$$G_{\xi,\sigma_u}(y) = \begin{cases} 1 - \left(1 + \frac{\xi}{\sigma_u} y\right)^{-\frac{1}{\xi}}, & \text{if } \xi \neq 0 \\ 1 - \exp\left(-\frac{y}{\sigma_u}\right), & \text{if } \xi = 0 \end{cases}$$

According to the sign of ξ, we have the following cases:

$$\xi > 0 : \text{distribution of Pareto type with heavy tail,}$$
$$\xi < 0 : \text{distribution of Beta type bounded above by } u - \frac{\sigma_u}{\xi},$$
$$\xi = 0 : \text{distribution of exponential type with light tail.}$$

1.2.2 Methods for Threshold Determination

The determination of the threshold is the most delicate step in the implementation of the POT approach, given that the quality of the model depends on it. The convergence of excesses to a GPD requires the determination of an appropriate threshold. The choice of threshold must be a compromise so that the determined threshold is sufficiently large to use asymptotic results, but not too high to obtain accurate estimates. However, choosing a low threshold can lead to uncertainties about the number of extreme observations and consequently produce biased estimates and a poor approximation of the asymptotic distribution. In this regard, several threshold determination methods are proposed in the literature, but in this part, we are going to present just one of this method.

Average Excess Function

We call the mean excess function also known as Mean residual life plot above a threshold u, the function $e(u)$ defined by:

$$e(u) = E[Y - u|Y > u]$$

with $0 \leq u \leq y_F$.
The mean excess can be used to guide the choice of the appropriate threshold u^*.
We have an important property: If $[Z = Y - u^*|Y > u^*] \sim G_{\xi,\sigma_{u^*}}$, the mean excess function $e(u)$ is linear in u for $u > u^*$.
The function $e(.)$ is a linear function of u. Therefore, the identification of the threshold u^* consists of searching for linearity on the graph defined by $(u, e(u))$ and called "Mean Residual Life Plot or Mean excess plot". In practice, the function $e(.)$ is approximated based on the empirical approximation $e_n(.)$:

$e_n(u) = \frac{\sum_{i=1}^{n}(Y_i - u)}{\sum_{i=1}^{n} \mathbb{1}_{\{Y_i > u\}}}$

This is the sum of excesses above the threshold u divided by the number N_u of data points that exceed the threshold u. If the tail of the distribution of X is heavy-tailed, then we observe that the mean excess function increases over time, whereas for light-tailed tails, it decreases over time. For instance, in the case of the Weibull distribution, as t approaches infinity, we have:

$$e(t) = t^{1-\tau}\frac{1}{\lambda\tau}(1 + o(1))$$

resulting in a mean excess function that ultimately decreases (respectively increases) if $\tau > 1$ (respectively $\tau < 1$). For a Pareto-type distribution, the mean excess function exhibits a linearly increasing behavior since when $\alpha > 1$, we have:

$$e(t) \sim \frac{t}{\alpha - 1} \text{ as } t \to \infty$$

, where α is the Pareto's parameter. Distributions with a finite endpoint x^+ demonstrate a mean excess function that ultimately decreases, and $e(x^+) = 0$.

1.2.3 Estimation of GPD Model Parameters

In this subsection, we focus on the different methods for estimating the parameter ξ or σ that are involved in the asymptotic distribution of extreme values. There are mainly two methods distinguished, namely parametric and non-parametric methods

a- Hill's estimator

Hill (1975) proposed the following estimator for the GPD distribution:

$$\hat{\xi}_k = \frac{1}{k-1}\sum_{i=1}^{k-1} \ln(X_{i,n}) - \ln(X_{k,n}),$$

where k is the highest order statistic (the number of exceedances), n is the sample size, and $\alpha = 1/\xi$ is the index of the distribution tail.

When plotting the Hill plot, it allows us to obtain estimates of the parameter α as a function of the highest order statistic (number of exceedances), thereby selecting the most stable index. The Hill

plot enables us to estimate the extreme quantile (threshold) using the tail index of the distribution found by the same method. Therefore, the Hill plot serves a dual purpose:
- Estimating the tail index of the distribution
- Estimating the threshold.

De Haan showed that the Hill estimator of the tail index converges in probability to its true value for any real α. The Hill estimator is more reliable in the domain of attraction of the Fréchet distribution, for which it provides a more efficient estimator of the tail index than other methods.

b- Maximum Likelihood Method

Consider a sample $Y_1, \ldots, Y_{k_n}$ iid with GPD distribution G_{ξ,σ_u}. The log-likelihood function is obtained from the GPD distribution, giving us:

$$\log\left(L(\xi, \sigma_u)\right) = -k_n \log(\sigma_u) - (1 + \frac{1}{\xi}) \sum_{i=1}^{k_n} \log\left(1 + \xi \frac{Y_i}{\sigma_u}\right),$$

when $1 + Y_i \frac{\xi}{\sigma_u} > 0$ for $i = 1, \ldots, k_n$, otherwise $(L_{MV}(\xi, \sigma_u)) = -\infty$.

The log-likelihood is maximized using numerical methods such as the Newton-Raphson algorithm. Maximum Likelihood Estimators are asymptotically Gaussian and efficient for $\xi > -\frac{1}{2}$. Furthermore, the Maximum Likelihood Estimator often encounters convergence and efficiency issues for small sample sizes ($n < 500$).

c- Method of Moments

The method of moments was introduced by Hosking and Wallis (1987) to estimate the parameters of the GPD distribution. The existence condition for the expectation and variance of a random variable Y with GPD distribution $G_{\xi,\sigma}$ is $\xi < \frac{1}{2}$.

For a GPD G_{ξ,σ_u}, estimation by the method of moments is based on the assumption that

$$E\left[(1 + \frac{\xi}{\sigma_u} Y)^r\right] = \frac{\sigma_u}{1 - r\xi}, \text{ if } 1 - r\xi > 0$$

In this case, we have:

$$\begin{cases} E(Y) = \frac{\sigma_u}{1-\xi} \\ V(Y) = \frac{(\sigma_u)^2}{(1-\xi)^2(1-2\xi)} \end{cases}$$

We can express the parameters of the GPD ξ and σ in terms of $E(Y)$ and $V(Y)$, so:

$$\begin{cases} \xi = \frac{1}{2}\left(1 - \frac{E(Y)^2}{V(Y)}\right) \\ \sigma = \frac{E(Y)}{2} \cdot \left(1 + \frac{E(Y)^2}{V(Y)}\right) \end{cases}$$

where $\bar{Y}$ and S_Y^2 are respectively the empirical estimators of the first and second moments of the sample. Thus, by replacing $E(Y)$ and $V(Y)$ with their respective empirical estimators, we obtain:

$$\bar{Y} = \frac{1}{k_n}\sum_{i=1}^{k_n} Y_i \quad \text{and} \quad S_Y^2 = \frac{1}{k_n - 1}\sum_{i=1}^{k_n}(Y_i - Y)^2,$$

and we obtain the estimators of $\hat{\xi}$ and $\hat{\sigma}_u$ defined respectively by:

$$\begin{cases} \hat{\xi} = \frac{1}{2}\left(1 - \frac{\bar{Y}^2}{S{*}_Y^2}\right) \\ \hat{\sigma_u} = \frac{1}{2}\bar{Y}\left(1 + \frac{\bar{Y}^2}{S{*}_Y^2}\right) \end{cases}$$

1.3 Goodness-of-Fit of Extreme Value Models

In practice, it is often necessary to analyze the validity of a model chosen for predictions. This involves ensuring that the selected models describe the studied data well and produce good estimates in order to provide more precision to the predictions and estimations.

1.3.1 Probability and Quantile Plot

The (P-P plot) and (Q-Q plot) are widely used to test the adequacy of extreme value models. These graphical tools allow for comparing the empirical distribution function for a (P-P plot) or the extreme quantiles (Q-Q plot) of a data sample to those from a theoretical distribution (such as GPD or GEV). When the observed distribution matches the theoretical distribution, the points on these plots align along the first bisector in the plane.

Suppose the observed values $y_{(1)}, \ldots, y_{(n)}$ are ordered in ascending order, a (P-P plot) is the set of points

$$\left\{\left(\hat{F}(y_{(i)}), \frac{i}{n+1}\right) : i = 1, \ldots, n\right\}$$

where $\hat{F}$ is the estimated empirical distribution and $i/(n+1)$ is the value taken by the estimated empirical distribution function in the interval $[y_{(i)}, y_{(i+1)}]$.

As for the (Q-Q plot), as its name suggests, it compares extreme quantiles. It consists of the set of pairs:

$$\left\{\left(\hat{F}^{-1}\left(\frac{i}{n+1}\right), x_{(i)}\right) : i = 1, \ldots, n\right\}$$

The following table summarizes the QQ plots of the most commonly used distributions in extreme value theory.

Distribution	$F(x)$	Coordinates
Normal	$\int_{-\infty}^{x} \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(u-\mu)^2}{2\sigma^2}\right) \mathrm{d}u$ $x \in \mathbb{R}; \mu \in \mathbb{R}, \sigma > 0$	$(\Phi^{-1}(p_{i,n}), x_{i,n})$
Log-normal	$\int_{0}^{x} \frac{1}{\sqrt{2\pi}\sigma u} \exp\left(-\frac{(\log u-\mu)^2}{2\sigma^2}\right) \mathrm{d}u$ $x > 0; \mu \in \mathbb{R}, \sigma > 0$	$(\Phi^{-1}(p_{i,n}), \log x_{i,n})$
Exponential	$1 - \exp(-\lambda x)$ $x > 0; \lambda > 0$	$(-\log(1 - p_{i,n}), x_{i,n})$
Pareto	$1 - x^{-\alpha}$ $x > 1; \alpha > 0$	$(-\log(1 - p_{i,n}), \log x_{i,n})$
Weibull	$1 - \exp(-\lambda x^{\tau})$ $x > 0; \lambda, \tau > 0$	$(\log(-\log(1 - p_{i,n})),$ $\log x_{i,n})$

1.3.2 Kolmogorov-Smirnov Goodness-of-Fit Tests

Let $x_1, \ldots, x_n$ be n realizations of a random variable X with a distribution function F. We are interested in determining whether it is reasonable to assume that X follows the distribution characterized by F, and we formulate the two test hypotheses:
1. H_0: X follows the distribution of F,
2. H_1: X follows a different distribution.
Thus, we use the empirical distribution function F_n of F defined by $F_n(x) = \frac{\mathrm{Card}(\{i | x_i \leq x\})}{n}$. The test statistic used is defined as:

$$D_n = \sup_{x \in \mathbb{R}} |F_n(x) - F(x)|.$$

We compare the computed value D_n to a critical value $D_\alpha(n)$ provided by the Kolmogorov-Smirnov tables. If $D_n > D_\alpha(n)$, we reject the null hypothesis H_0 with a risk of error α.

II Estimating of reinsurance premiums

In the field of reinsurance, estimating the pure premium is crucial. It relies on the assumption that the total cost of claims, represented by the non-negative random variable X, follows a survival function $\overline{F}(x)$, defined as the probability that claims exceed a certain amount x. The calculation of the premium is based on the net premium principle, which states that if an insurer sells a sufficient number of independently and identically distributed policies, the risk becomes negligible. Thus, on average, the insurer will not incur losses when using this premium. The pure premium, denoted as Π, equals the mathematical expectation of X, which is the integral of the survival function over all possible values of x:

$$\Pi = E(X) = \int_0^{\infty} \overline{F}(x)\, dx$$

where $\overline{F}(x)$ represents the probability that X exceeds x.

Or

$$\Pi = \mathbb{E}(X) = \int_0^{x_F} \overline{F}(x)\, dx, \quad \text{where } x_F = \sup\{x : F(x) < 1\}$$

This value of Π is interpreted as the premium that enables the company not to lose money when it sells independent and identically distributed policies.

2.1 Wang's premium principle

The Wang's premium principle provides a new estimate of the premium Π using a transform of the survival function $\overline{F}(x)$ by an increasing, concave function noted g.

In this case, the expression of Π is now given by :

$$\Pi = E(X) = \int_0^{\infty} g(\overline{F}(x))\, dx$$

The most common function g used are the following :

1. Net Premium Principle:

$$g(x) = x \implies \Pi = \mathbb{E}(X)$$

2. Proportional Hazard Transform Principle:

$$g(x) = x^{1/\alpha} \quad (\alpha > 1)$$

3. Dual-Power Transform Principle:

$$g(x) = 1 - (1-x)^{\alpha} \quad (\alpha > 1)$$

4. Gini Principle:

$$g(x) = (1+\alpha)x - \alpha x^2 \quad (0 < \alpha < 1)$$

5. Square Root Function Principle:

$$g(x) = \frac{\sqrt{1+\alpha x} - 1}{\sqrt{1+\alpha} - 1} \quad (\alpha > 0)$$

These different functions of g express transformation principles used in various fields, such as insurance and finance, to model costs, risks or other relevant variables. Each function embodies a distinct principle, offering a unique approach to characterizing the relationships between variables, and thus enabling specific analyses and calculations. By selecting the appropriate function according to context, it is possible to better understand the underlying dynamics and make informed decisions on premium pricing, risk management or other relevant applications.

2.2 Excess Loss Reinsurance with a Global Deductible

Consider now an excess of loss reinsurance with an overall deductible R. The survival function of the total cost $(X-R)^+$, where $y^+ = \max(0, y)$, is $\overline{F}(x+R)$ where $x > 0$. Therefore, the pure premium $\Pi(R)$ is given by

$$\Pi(R) = \mathbb{E}((X-R)^+) = \int_R^{\infty} \overline{F}(x)\, dx = \int_R^{\infty} (1 - F(x))\, dx.$$

The Π formula for any function g of Wang's principle is given by :

$$\Pi = \int_R^{\infty} g(\overline{F}(x))\, dx$$

The pure premium Π is determined by integrating a function $g(\overline{F}(x))$ over the range of claims exceeding the deductible R to infinity. Here, $\overline{F}(x)$ represents the decumulative distribution function of the original claim amount, and $g(\cdot)$ is a function used to adjust the premium for claims within that layer.

In the realm of reinsurance, there's a notable focus on understanding extreme occurrences—those events with infrequent but significantly impactful outcomes. Extreme value statistics emerge as a vital tool for analyzing such rare events, particularly emphasizing the tail behavior of distributions, often described by the extreme value index. Within this theoretical framework, various estimators are employed, including the extreme value index and small exceedance probabilities, which play a crucial role in determining reinsurance premiums.

2.3 Karamata theorem for estimate the premium

If we assume that $G(t) = g\left(\frac{1}{t}\right)$ is such that $G(t) = t^{\beta} \cdot L'(t)$, where $L'(t)$ is slowly varying at infinity, then we can apply Karamata's theorem.

we already know that

$$\overline{F}(x) = x^{-1/\xi} L(x)$$

Then

$$\Pi(R) = \int_R^{\infty} g(\overline{F}(x))\, dx = \int_R^{\infty} G\left(\frac{1}{\overline{F}(x)}\right)\, dx$$

The final expression for Π is of the form:

$$\Pi(R) = \int_R^{\infty} e^{\frac{\beta}{\xi}} \cdot L''(x)\, dx$$

where L" also slowly varying at infinity

Karamata's inequality states that

$$\xi < -\beta.$$

According to Karamata's theorem, Π is equivalent to $\tilde{\Pi}$ which is written as follows:

$$\Pi(R) \sim \tilde{\Pi}(R) = \frac{1}{-\beta/\xi - 1} R g(\overline{F}(R))$$

when the retention level R tends to infinity.

The work on extreme value statistics and reinsurance premium estimation represents a significant advancement in understanding and applying these crucial concepts. By precisely defining key terms such as the survival function $G(t)$ and the regular variation index β, a solid foundation is laid for the subsequent analysis. Applying Karamata's theorem and developing an asymptotic expression for the gross premium $\tilde{\Pi}(R)$ as the retention level R tends to infinity demonstrates a deep understanding of the underlying theoretical concepts and their practical application in reinsurance premium estimation. The clarity established in the relationship between net and gross premiums, represented by $\tilde{\Pi}(R)$ and $\Pi(R)$ respectively, provides crucial insights into the pricing process in the insurance industry.

We can also estimate Π by $\hat{\Pi}(R) = \frac{1}{-\beta/\hat{\xi}-1} R_g(\hat{p}_R)$

where $\hat{\xi}$ is the estimator of ξ and $\hat{p}_R$ is the estimator of $p_R = \overline{F}(R)$.

2.4 Expression of premium estimation using GPD, EPD and Pareto models

We saw earlier the expression of Π which is of the form for :

$$\Pi(R) = \int_R^\infty g(\overline{F}(x))\,dx$$

And if we take g=id (Net premium principle), the formula becomes :

$$\Pi(R) = \mathbb{E}((X-R)^+) = \int_R^\infty \overline{F}(x)\,dx = \int_R^\infty (1-F(x))\,dx.$$

The estimated Π noted $\hat{p}_R$ formulas for the various are given by :

2.4.1 Estimaton of excess-loss premiums using GPD model

Π estimate:

$$\hat{\Pi}(R) = \frac{k+1}{n+1} \cdot \frac{\hat{\sigma}_k}{(1-\hat{\gamma}_k)(1+\frac{\hat{\gamma}_k}{\hat{\sigma}_k}(R-X_{n-k,n}))^{\frac{1}{\hat{\gamma}_k}-1}}$$

with $\hat{\sigma}_k$ and $\hat{\gamma}_k$ the estimates for the parameters of the GPD.

2.4.2 Estimaton of excess-loss premiums using EPD model

The Extended Pareto Distribution (EPD) is an extension of the classical Pareto distribution. It incorporates an additional shape parameter, often denoted as τ, allowing for more flexibility in modeling tail behavior. This extension enables the distribution to capture a wider range of tail behaviors, including heavier or lighter tails compared to the classical Pareto distribution.

The estimation of the premium with EPD model is Π:

$$\hat{\Pi}(R) = \frac{k+1}{n+1} \times X_{n-k,n}^{\frac{1}{\hat{\gamma}}} \times \left(\left(1-\frac{\hat{\kappa}}{\hat{\gamma}}\right) \times \left(\left(\frac{1}{\hat{\gamma}}-1\right)^{-1} \right) \times R^{1-\frac{1}{\hat{\gamma}}} + \frac{\hat{\kappa}}{\hat{\gamma} \times X_{n-\hat{k},n}^{\hat{\tau}}} \times \left(\left(\frac{1}{\hat{\gamma}}-\hat{\tau}-1\right)^{-1} \right) \times R^{(1+\hat{\tau}-\frac{1}{\hat{\gamma}})} \right)$$

with $\hat{\gamma}$, $\hat{\tau}$ and $\hat{\kappa}$ the estimates for the parameters of the EPD.

2.4.3 Estimaton of excess-loss premiums using Pareto model

The estimated Π is given by:

$$\hat{\Pi}(R) = \frac{k+1}{n+1} \cdot \left(X_{n-k,n}^{\frac{1}{H_{k,n}}} \cdot R^{1-\frac{1}{H_{k,n}}} \right) \div \left(\frac{1}{H_{k,n}} - 1 \right)$$

with $H_{k,n}$ the Hill estimator.

2.5 Non parametric approach

By setting the retention threshold to R, to estimate the pure premium, we can consider the empirical mean of $E[(X-R)_+]$, where X denotes the variable representing claim amounts. The empirical mean is thus given by:

$$\frac{1}{n}\sum_{i=1}^{n}(X_i-R)_+$$

III Case study

For the code, please refer to the HTML file that has been included in the folder.

3.1 About the ReIns package

The ReIns package is an R package that contains many useful libraries for the application of Extreme Value Theory (EVT). These libraries provide graphical tools and pre-implemented estimators in the R language that allow for the estimation of parameters of certain distributions suitable for modeling extremes. This package also provides databases (real and simulated) suitable for the application of Extreme Value Theory. These data mainly come from the insurance world, including:

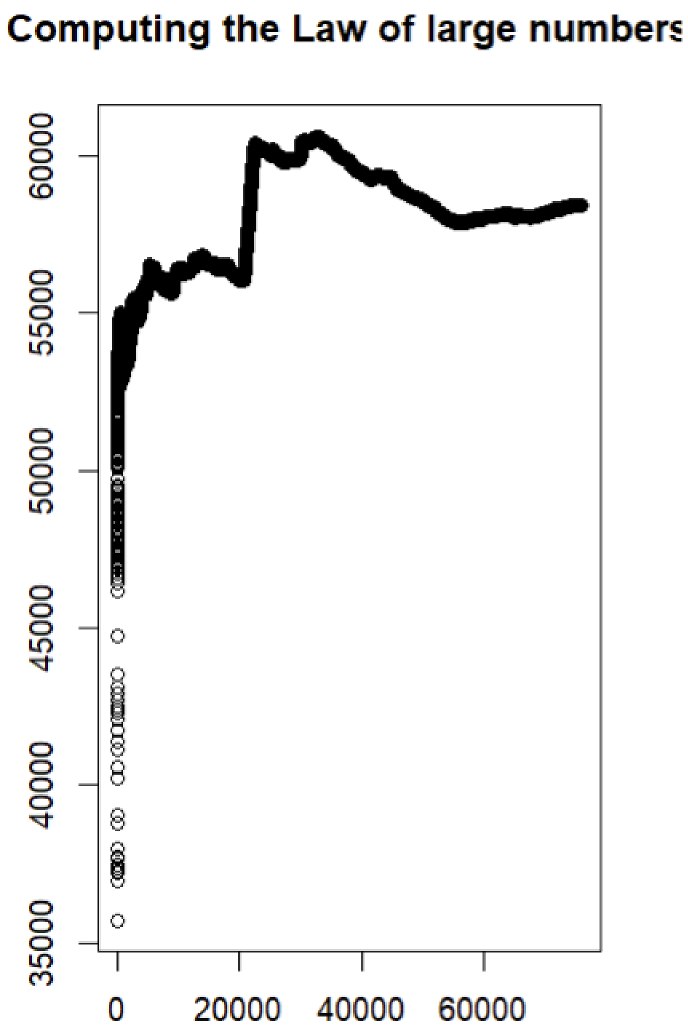
- Norwegian fire insurance data (`norwegianfire`): This dataset contains information related to a Norwegian insurance company, consisting of 9,181 rows and two columns (year of claim, claim amount ranging from 500 to 465,365 NOK).
- SOA group medical insurance data (`soa`): This dataset contains medical claim costs for the year 1991 in the USA. All claims exceed $25,000 USD.It contains one column (size of the claim) and 75,789 rows. In order to make easy the implementations, we decided to work with only the last 500 high amounts.
- Secura Re automobile reinsurance data (`secura`): This dataset contains claims for the period 1988-2001 from multiple European insurance companies. All claims are above 1,200,000 euro.

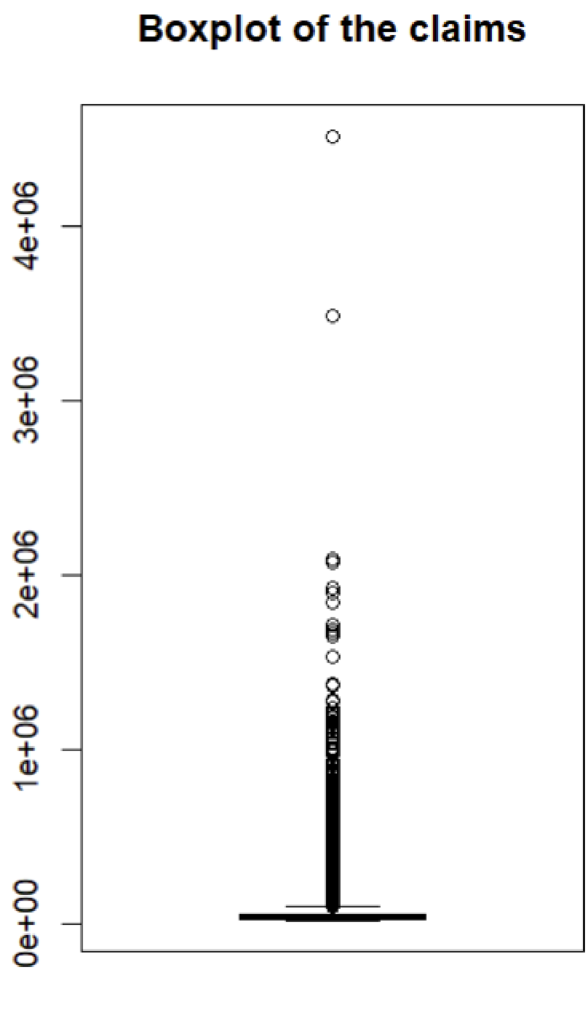
In this section, we will apply the entire theory as presented above. We will start by performing a graphical analysis, selecting the distribution that best fits our data, and estimating its parameters. We will then estimate quantiles and other important information for an actuary.

3.2 Graphical analysis

Based on the statistical summary below, we suspect that there are several values that deviate too much from the mean. To ensure that we can apply extreme value theory, we inspect the boxplot and empirically represent the sum involved in the law of large numbers.

Min	25,000
1st Qu.	30,543
Mean	58,413
Median	40,224
3rd Qu.	61,322
Maximum	4,518,420

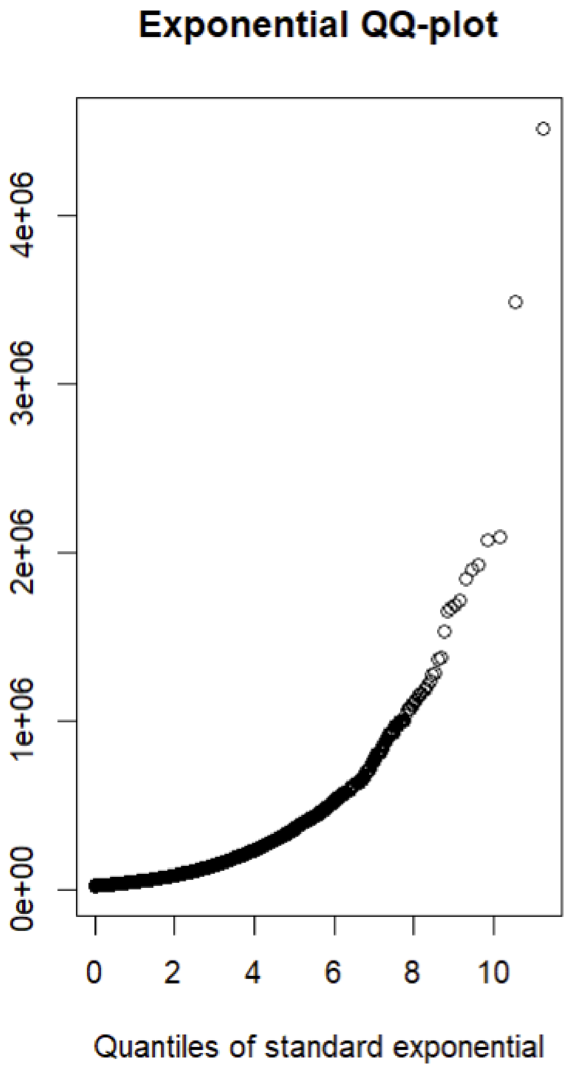


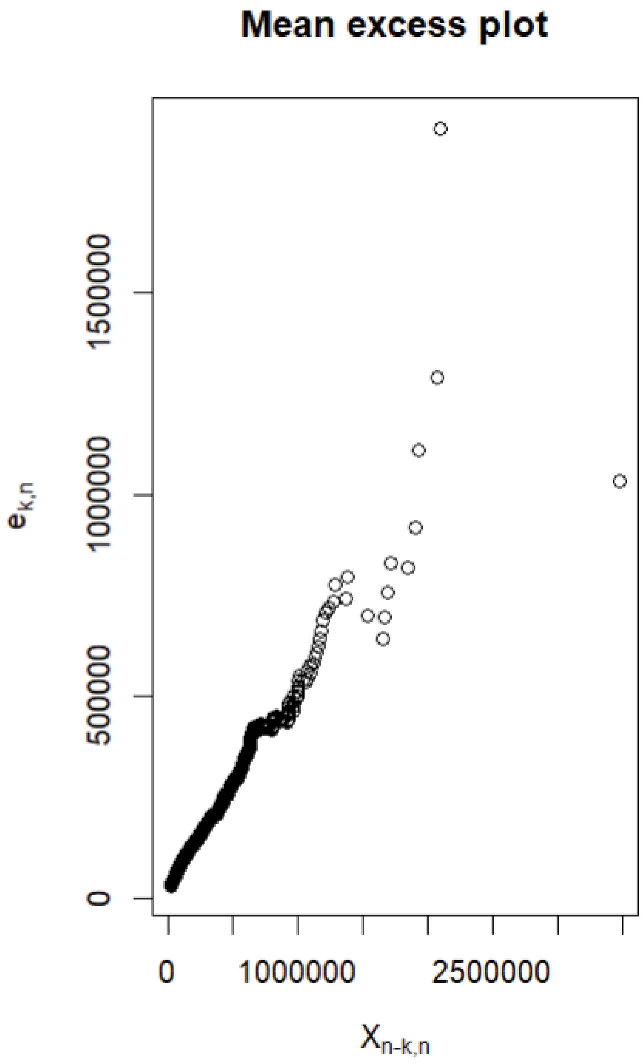


By analyzing the implementation of the law of large numbers, we observe that the mean does not appear very clearly even with 20,000 observations. We observe significant fluctuation, which is somewhat concerning. To reinforce our intuitions, we explore the boxplot of the data and observe that there are many values that deviate from the mean. We notice numerous outliers and many large values. An approach using the extreme value method seems appropriate.

As mentioned earlier, among the important tools for studying extremes is the Mean Excess. It helps determine whether our data has Heavy Tails (HT) or Light Tails (LT). It is represented jointly with the exponential quantile quantile quantile plot.

First, we represent the exponential quantile quantile plot of the data, in order to determine if the data come from HT or LT.

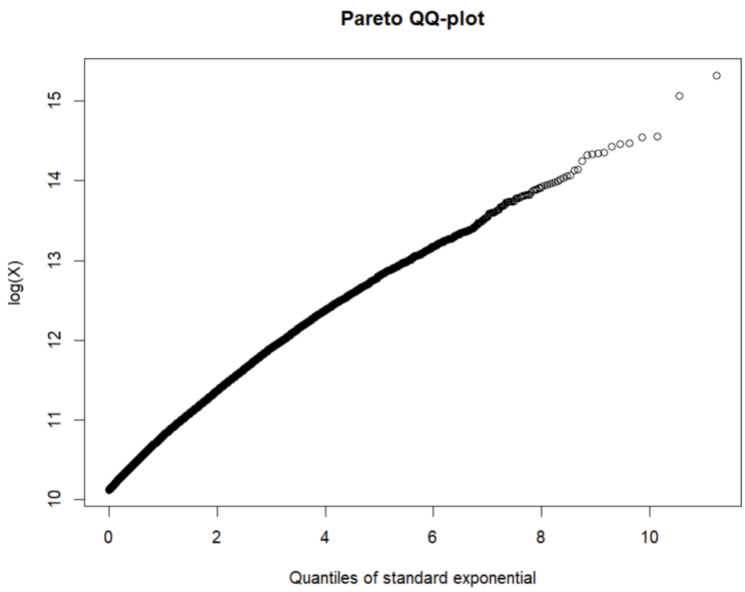




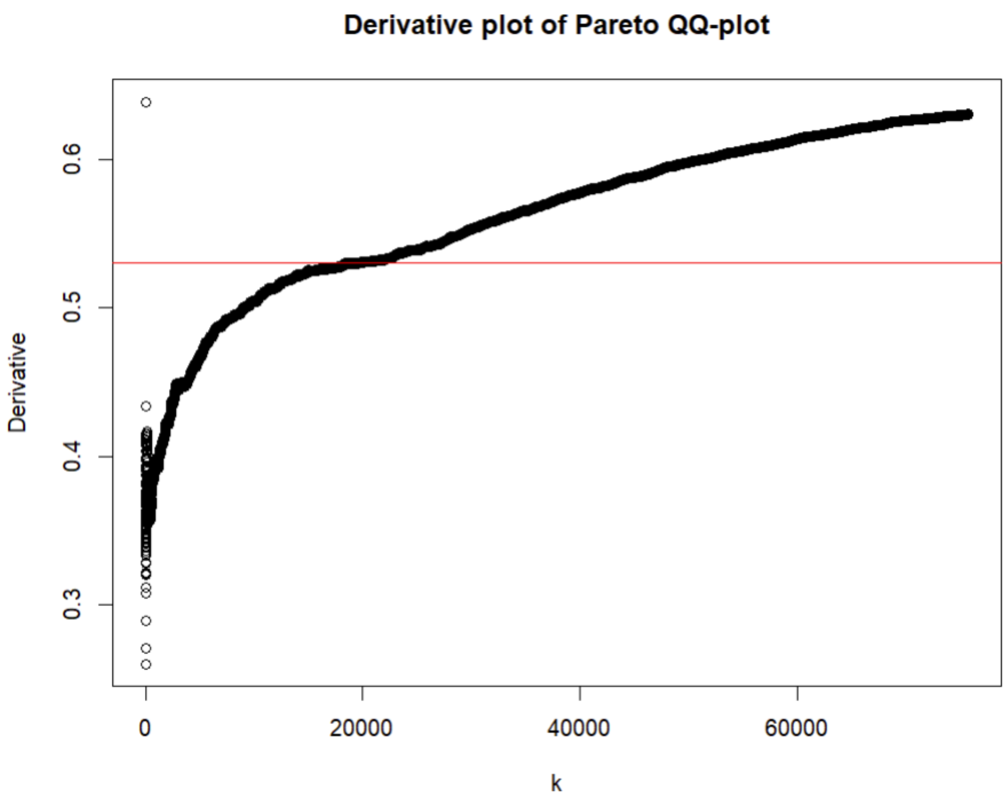
We can see the that the exponential quantile quantile plot is convex, and increase with $X_{n-k,n}$, this means that our data come from an heavy tail distribution.

We also represent the Mean Excess, which is the derivative quantile plot (thus should follow a diagonal). In our case, it is linear for a moment and increases according to $X_{n-k,k}$ (thus decreasing according to k). An approximation by a Pareto distribution may be appropriate.

When modeling reinsurance claim data, we anticipate convex exponential QQ-plots linked with increasing mean excess plots $(X_{n-k,n}, e_{k,n})$. A common subsequent step is to examine log-normal or Pareto QQ-plots. It is worth noting that the mean excess plots of a Pareto-type distribution will ultimately be linearly increasing with a slope of $1/(\alpha - 1)$, where α is the parameter of a strict pareto distribution. By examining the Mean Excess, which is linear from a certain point onwards, we can infer that the Pareto distribution would fit our data well. To confirm this, we plot the Pareto QQ plot.

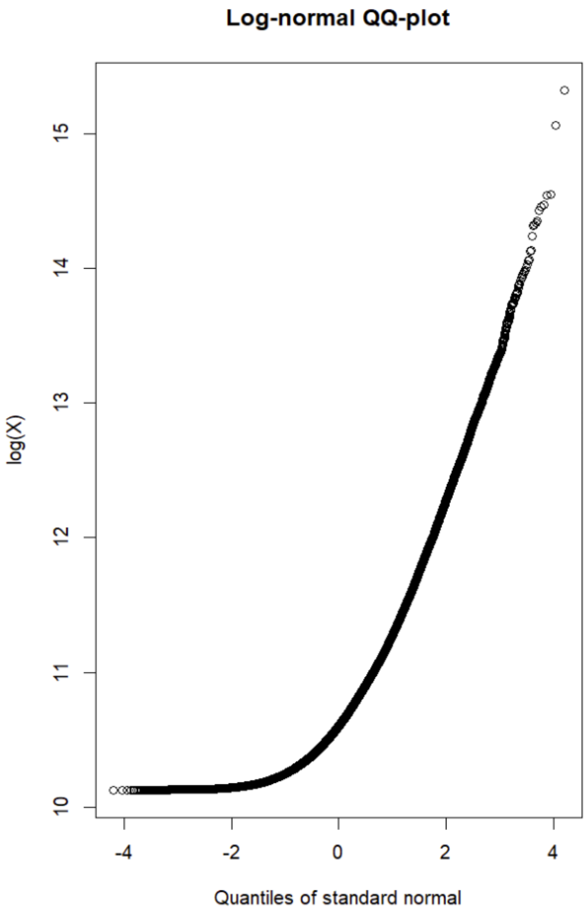


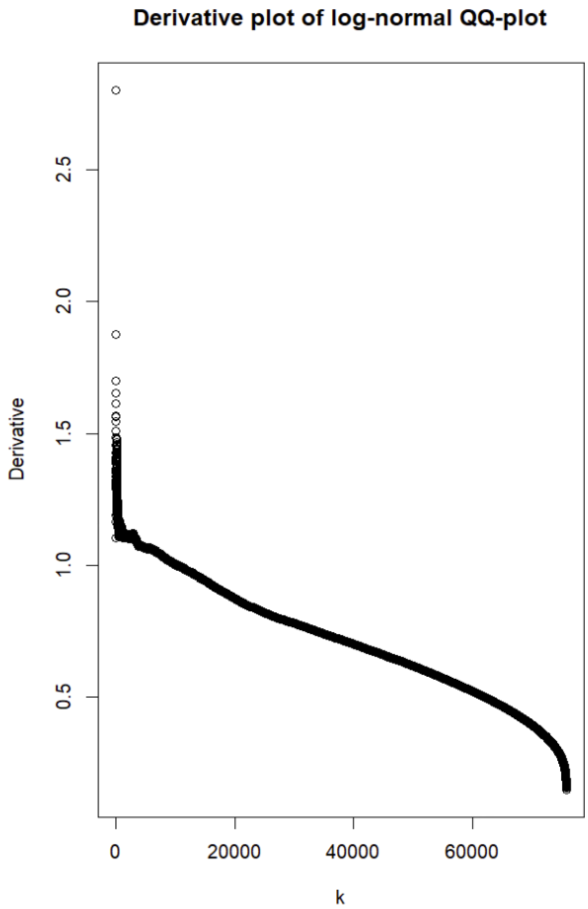
The Pareto QQ-plot is linear, indicating that the Pareto distribution is suitable. Our intuition is thus confirmed. The derivative plot can be utilized to estimate the tail index $\gamma = \frac{1}{\alpha}$ of the Pareto distribution. It's worth noting that these derivatives are essentially the Hill estimates.



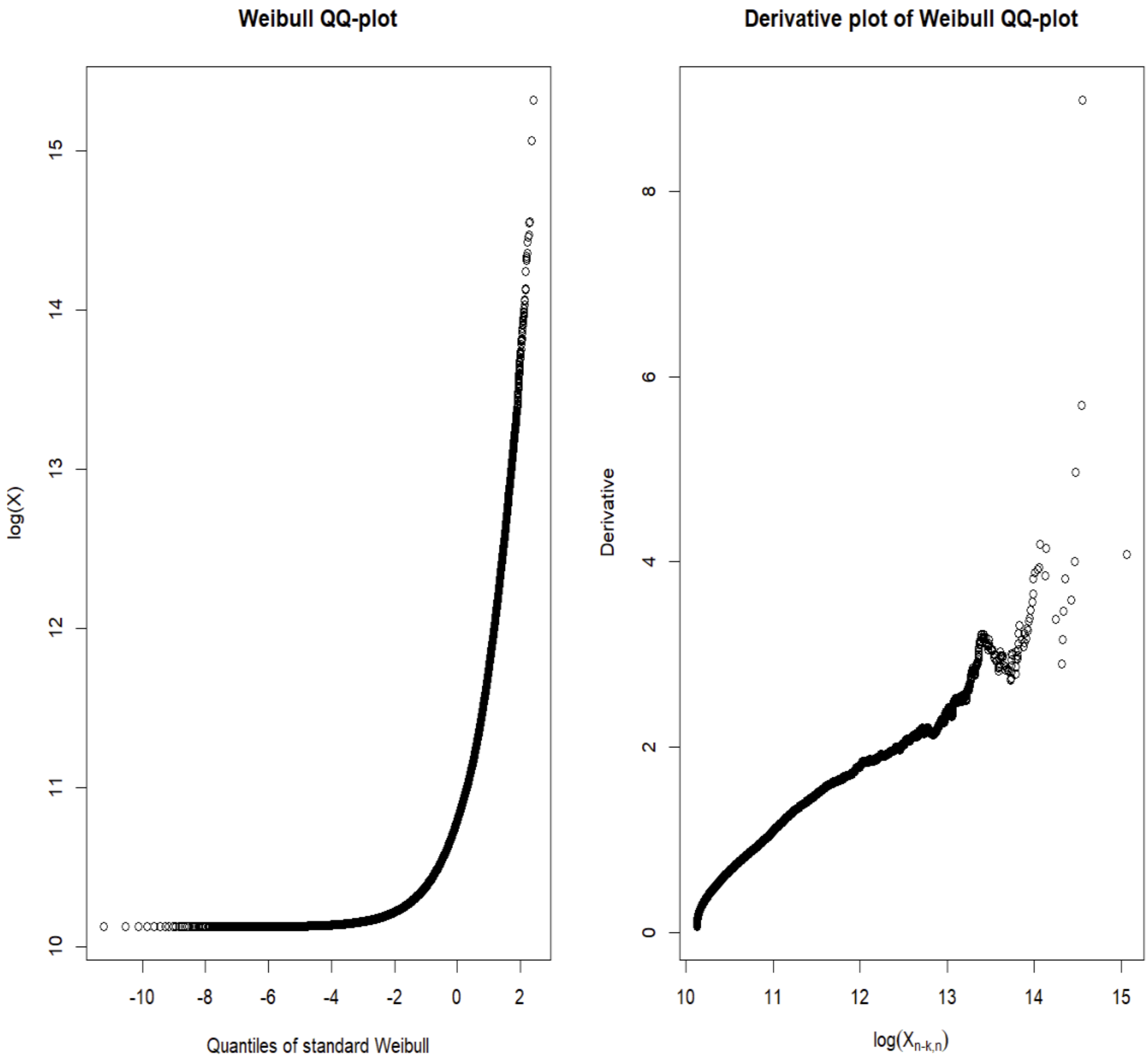
When looking at the Pareto QQ derivative plot, a horizontal line (thus a certain stablity) can be observed for $\gamma \approx 0.53$ with $k \in [18,000; 24,000]$.

We also look to the log normal QQ plot and its derivative.





The log-normal QQ plot does not exhibit a line of the form $y = x$. Thus, the log-normal distribution is not suitable for our data. We then proceed to test the Weibull distribution, as suggested in the book "Reinsurance: Actuarial and Statistical Aspects" by Hansjörg Albrecher.

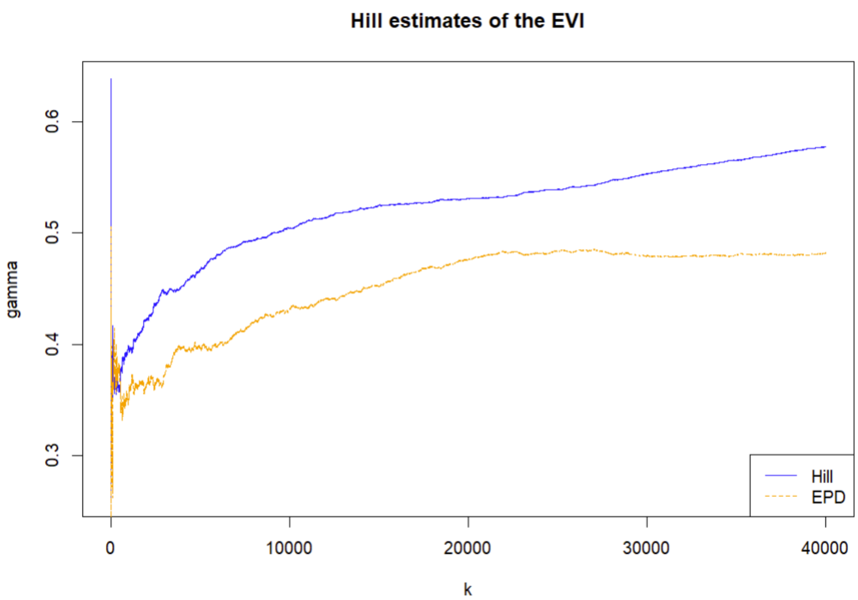


It is clear that weibull distribution does not fit correctly our data. Finally, we will work with the pareto distribution.

3.3 Some estimations

3.3.1 Estimating a positive Extreme Value Index

As told in precedent sections, the Extreme Value Index (EVI) can be obtained thanks to Hill estimator, by fitting a (strict) pareto distribution. In the previous sections, the EVI was noted ξ, in R it is noted γ. We use the command *Hill*. This command produces a plot that helps determine the appropriate value for γ. The bias of the Hill estimator can be problematic.One solution is to use the EPD estimator, which is for the Extended Pareto distribution instead of the (strict) Pareto distribution and which helps reducing the bias of the Hill estimator.

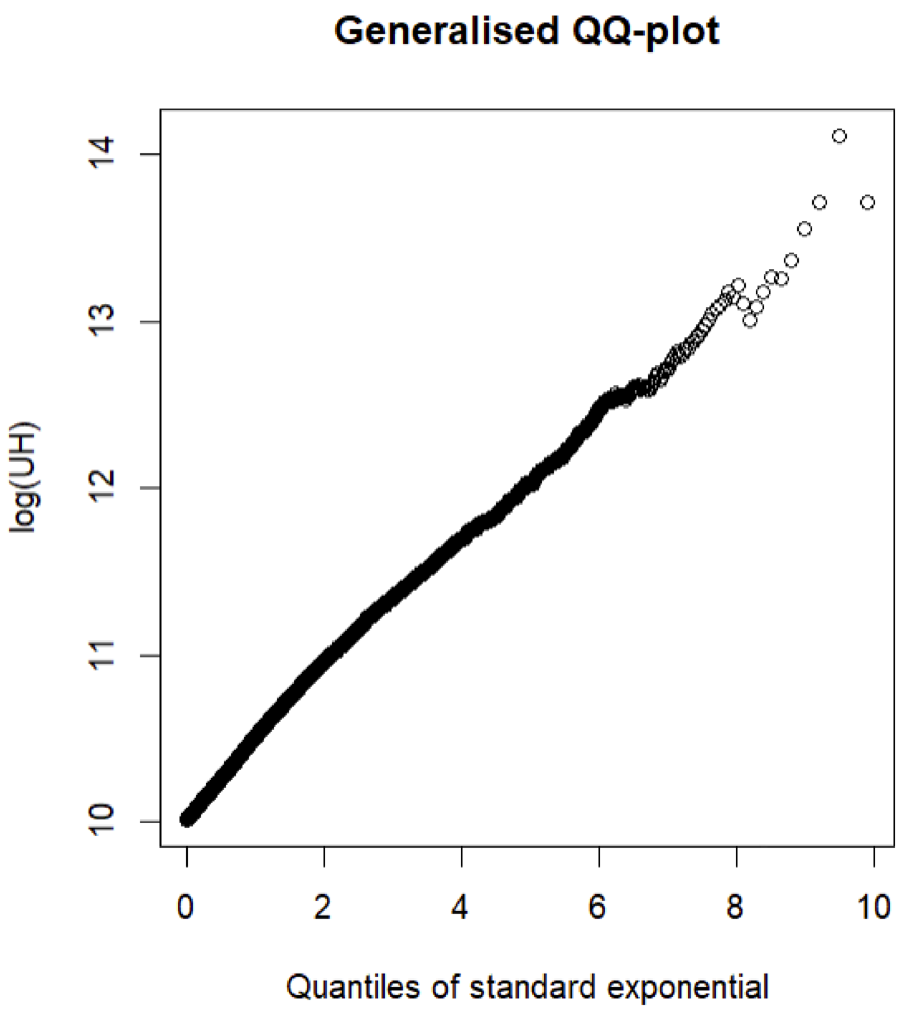


To determine the optimal choice of k, we examine the points where the two plots intersect (avoiding too low points, as high variance may result).

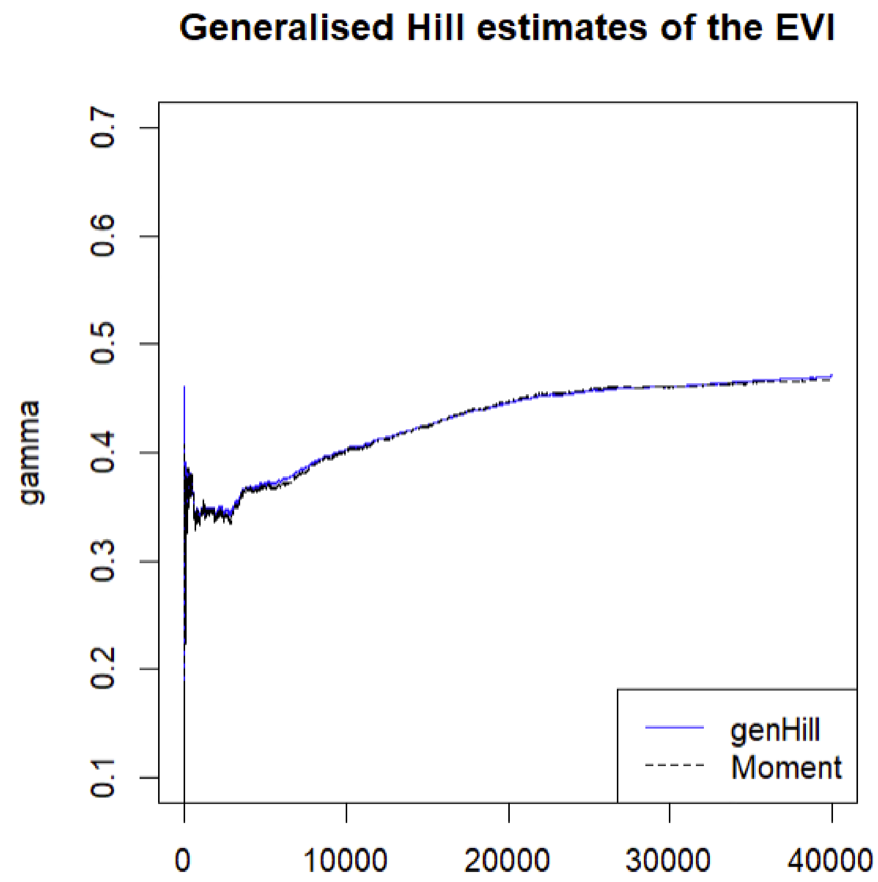
On the graph, we observ $k \approx 3,000$ and $\gamma \approx 0.44$. γ is potitive, our dataset has an Heavy Tail, and belongs to the frechet domain.

3.3.2 Estimating Extreme Value Index whatever its sign

In the precedent section, the presented methods are used only to estimate a positive EVI. To estimate a negative or null EVI, one can use the *genQQ* command available in R. Note that the EVI is nothing more than the slope of the GenQQ.

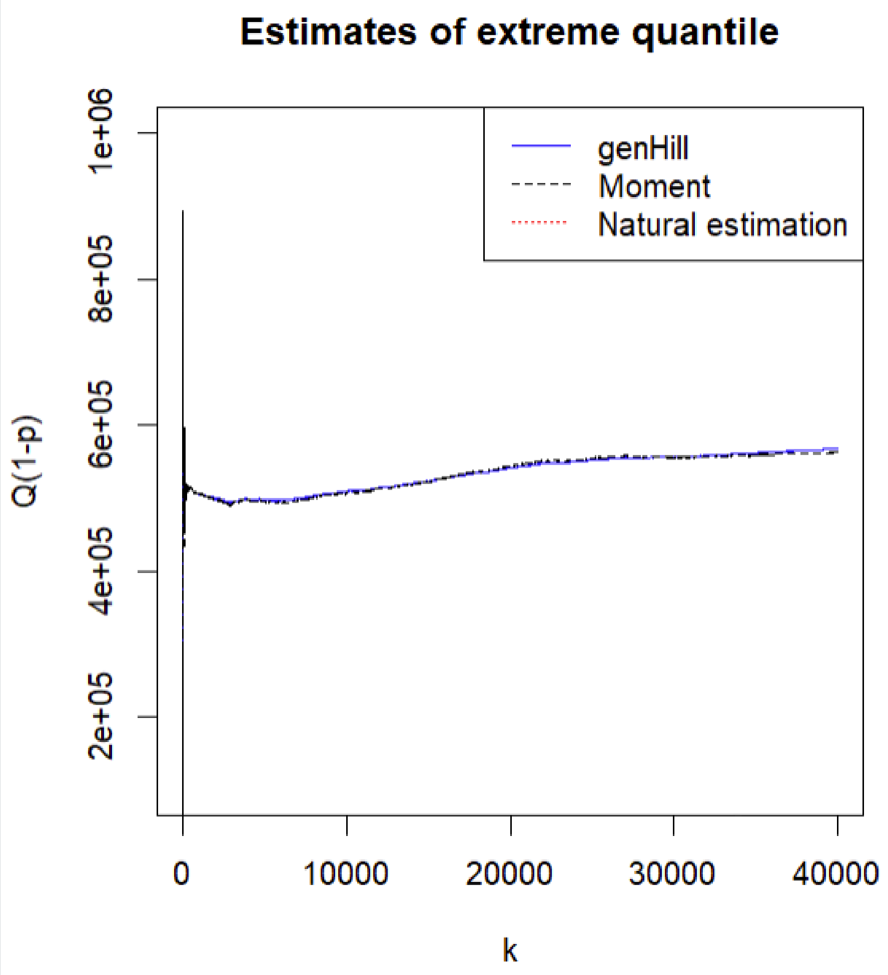


Estimating the slope gives $\gamma \approx 0,48$. One can also use the generalised Hill estimator, which estimates directly the slope of the Generalised QQ plot above. Another method is to consider the moment estimator.



3.3.3 Quantiles estimation

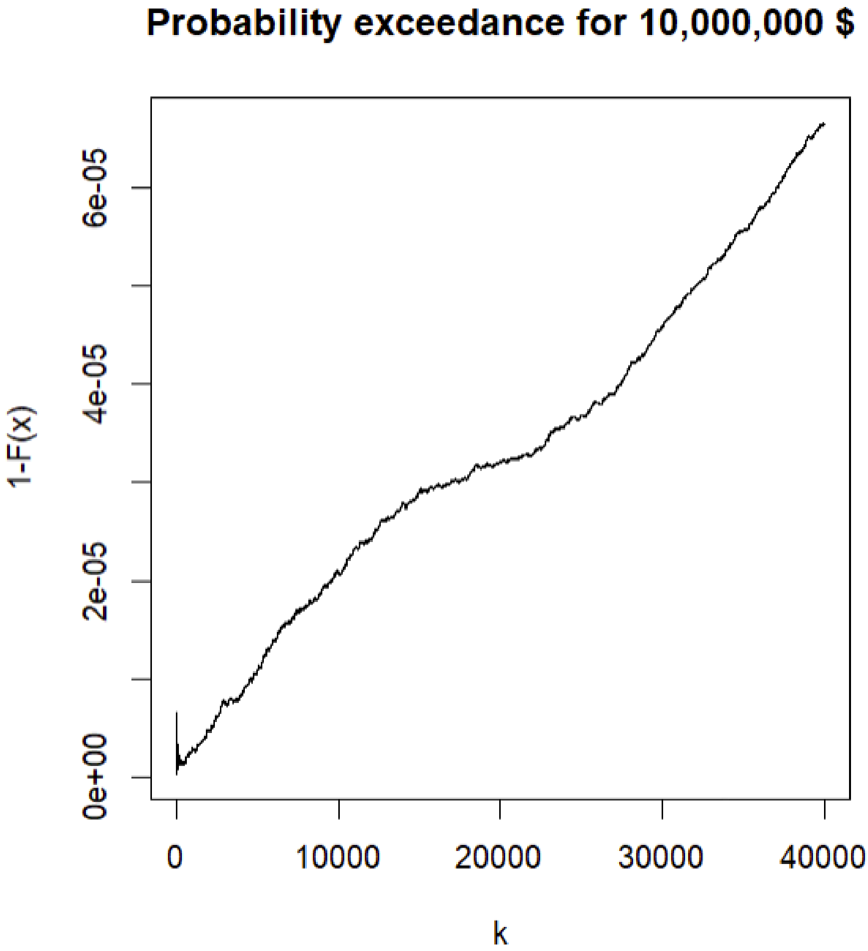
Quantiles play a crucial role in insurance as they provide valuable insights into the distribution of potential losses or claims. By analyzing various quantiles such as the median, lower quantiles (e.g., 5th or 10th percentiles), and upper quantiles (e.g., 95th or 99th percentiles), insurers can assess the likelihood and severity of different scenarios. Quantiles help in setting appropriate reserves, determining premium rates, and managing risk effectively in insurance operations. Moreover, quantile-based risk measures such as Value at Risk (VaR) and Conditional Tail Expectation (CTE) are widely used for quantifying and managing extreme risks in insurance portfolios. Insurance companies must have a bankruptcy risk of 0.5%. In our case, how much money does the company need to meet this criterion? To do this, we can use quantile estimation by the method of moments and by inversing the cumulative distribution function of the Pareto distribution that best fits our data.



The Value at Risk Var(99,5%) $\approx$ 500,000$ US

3.3.4 Small exceedance probabilities and return period

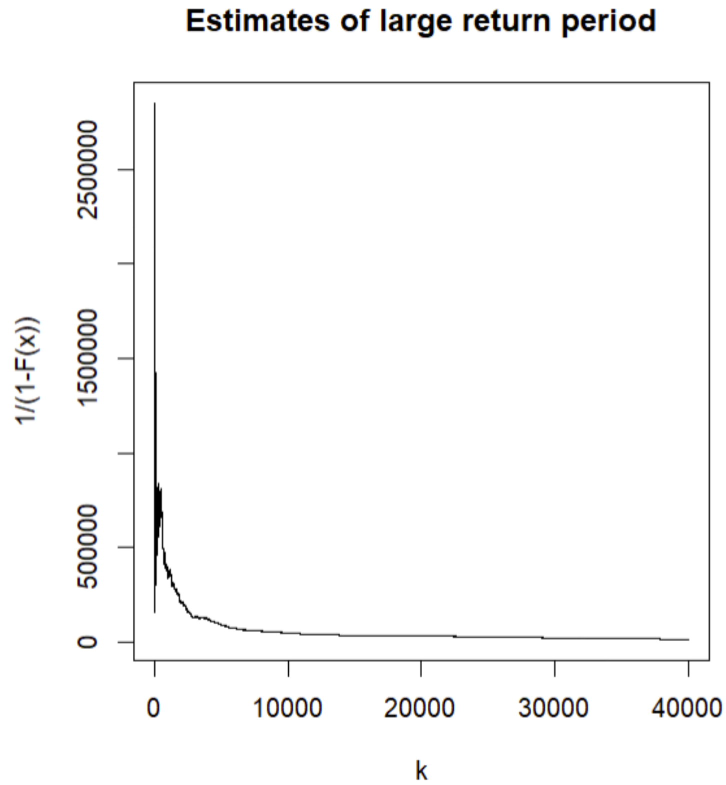
Now we will focus on estimating small probabilities $P[X > q]$, and return periods obtained by $1/P[X > q]$. For illustrations, we choose $q = 10,000,000\$$.



In $III.3.1$, we proved that the good choice of k was around $3,000$.

For $k \approx 3,000$, the estimated probability is around 10^{-5}. Then the return period should be around 100,000 (claims).

On the graph below, and for $k \approx 3,000$, we obtain a return period $\approx$ 100,000 (claims).



3.3.5 Premiums in an XL treaty

Reinsurance is a risk management strategy used by insurance companies to transfer a portion of their risks to other insurers, known as reinsurers. In reinsurance arrangements, the reinsurer agrees to indemnify the insurer for a portion of the covered losses in exchange for a premium. This allows insurers to mitigate their exposure to large and unexpected losses, maintain solvency, and stabilize financial performance. Reinsurance plays a crucial role in the insurance industry by providing additional capacity, spreading risks globally, and enhancing insurers' ability to underwrite large risks or catastrophic events. Moreover, reinsurance facilitates innovation, promotes competition, and contributes to the overall stability of the insurance market.

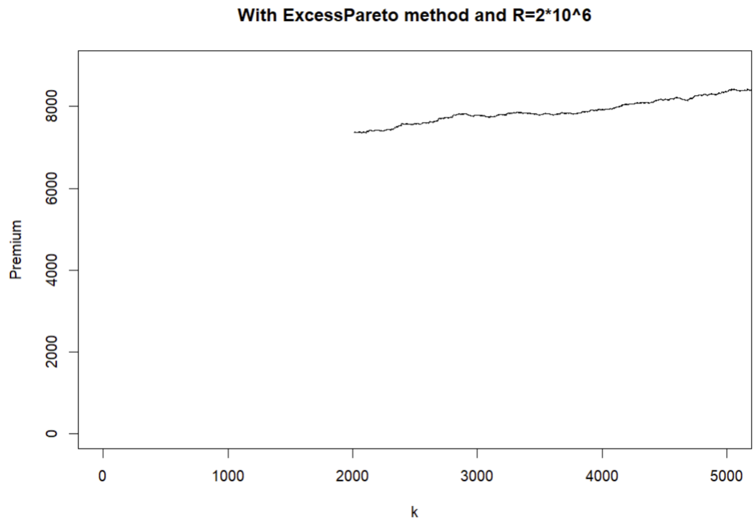
There are several types of reinsurance treaties. In our work, we will focus more on the XL (excess-loss) treaty.

Excess-Loss (XL) Treaty: The XL treaty is a type of reinsurance agreement designed to protect the insurer against large and catastrophic losses that exceed a predetermined threshold, known as the retention limit. Under the XL treaty, the reinsurer agrees to indemnify the insurer for losses that exceed the retention limit up to a certain coverage level, typically referred to as the excess layer. XL treaties are often used to cover high-severity, low-frequency events such as natural disasters or major accidents. By providing coverage for catastrophic losses, XL treaties help insurers manage their exposure to extreme risks and maintain financial stability.

In the following sections, we will explore how to estimate the pure premium of an XL treaty.

Pure Premium Estimation

With the precedent section 2.5, we can estimate premium for R=200,000 $



The beginning of this part showed us that the optimal choice of k was $\approx 3,000$. On the graph above, the premium is approximatively 7,000 \$. With the non parametric approach, we obtain a premium 6903.799 \$.

We did'nt use the ExcessGPD approach, because it took too long to find the MLE parameters for the computer.

Retention level R	**Non Parametric approach**	**Excess-pareto approach**
500,000	1,525.17	$\approx 1,800$
700,000	860.5265	$\approx 1,000$
900,000	540.8776	≈ 700
1,000,000	436.6941	≈ 500

Conclusion

In insurance, understanding rare phenomena (with low probability of occurrence but high intensity) is crucial for insurers. Extreme Value Theory (EVT) provides an appropriate framework for studying such phenomena, modeling and managing rare but potentially devastating events, thanks to valuable tools. This research work has allowed us to gain a better understanding of EVT and has provided us with the necessary basics to manipulate data suitable for this type of analysis. It has also enabled us to comprehend the importance of reinsurance. Indeed, insurance companies are required to maintain a certain solvency ratio, defined by the ACPR (Autorité de Contrôle Prudentiel et de Résolution), and the occurrence of extreme claims can jeopardize the solvency of the insurance company, potentially leading to bankruptcy.

Appendix

For this study and research work, we first did a document search on the extreme values theory. We first focused on the theory and our supervisor Ms. Masiello helped us understand it. We then thought about the applications to be made in this TER. After several discussions with Ms. Masiello as well as a lot of personal research, we had to select the applications that seemed most relevant and not too difficult to do. Our group then came together to set up the coding on the R software, particularly on the pricing work. Once the applications are completed, we distribute the tasks for writing the report. Julien MARI took care of writing the elements of extreme value theories. Abibou FALL designed his theoretical part on pricing. Zié COULIBALY wrote all the interpretations of the applications previouslycarriedout.

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